

AbstentionBench: Reasoning LLMs Fail on Unanswerable Questions

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For Large Language Models (LLMs) to be reliably deployed in both everyday and high-stakes domains, knowing when *not* to answer is equally critical as answering correctly. Real-world user queries, which can be underspecified, ill-posed, or fundamentally unanswerable, require LLMs to reason about uncertainty and selectively *abstain*—i.e., refuse to answer definitively. However, abstention remains understudied, without a systematic evaluation framework for modern LLMs. In this work, we introduce **AbstentionBench**: a large-scale benchmark for holistically evaluating abstention across 20 diverse datasets, including questions with unknown answers, underspecification, false premises, subjective interpretations, and outdated information. Evaluating 20 frontier LLMs reveals abstention is an unsolved problem, and one where scaling models is of little use. While recent reasoning LLMs have shown impressive results in complex problem solving, surprisingly, we find that reasoning fine-tuning *degrades* abstention (by 24% on average), even for math and science domains on which reasoning models are explicitly trained. We find that while a carefully crafted system prompt can boost abstention in practice, it does not resolve models’ fundamental inability to reason about uncertainty. We release **AbstentionBench** to foster research into advancing LLM reliability.

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Code: <https://github.com/facebookresearch/AbstentionBench>



1 Introduction

Reliability is key to user trust in Large Language Models (LLMs). If users can’t trust model responses, we can’t fully benefit from their application—in either everyday or high-stakes settings [1, 2, 3]. However, faced with a changing world and noisy, ambiguous, or unanswerable user queries, there will always be cases where a reliable response is impossible: models need not only answer with high accuracy, but must also know when *not* to answer. For example, the answer to the important query “My dog was prescribed 5mg/kg Prednisone, how much should I give her?” depends on the specific dog’s weight, here left unspecified. Reliable models must recognize such uncertainty and *abstain*—i.e., avoid providing a definitive answer—instead expressing uncertainty, clarifying, or simply responding “I don’t know”. To do this successfully, LLMs need to reason about both evidence and uncertainty, weighing the information available to determine whether an answer is appropriate.

While in traditional machine learning classification, abstention approaches rely on the well-defined notions of aleatoric (inherent randomness) or epistemic uncertainty (limited training data), abstention in LLMs is more complex. Given the open-ended nature of LLM dialogue, LLMs must be able to abstain faced with a wide range of user queries, ranging from vague, underspecified questions, through those with no known answer, to those based on false premises. Previous research has predominantly studied LLM uncertainty and refusal in the context of safety, factuality, and hallucination [4, 5, 6], neglecting other diverse abstention scenarios. While individual datasets have evaluated abstention in isolated contexts, there is no holistic benchmark for comprehensively evaluating abstention.

In this paper we introduce **AbstentionBench**—a benchmark for evaluating the ability of LLMs to abstain under uncertainty (Fig. 1). We conduct a systematic review of datasets related to abstention and curate 17 high-quality datasets spanning 6 diverse scenarios. To extend our analysis to reasoning-heavy domains, we

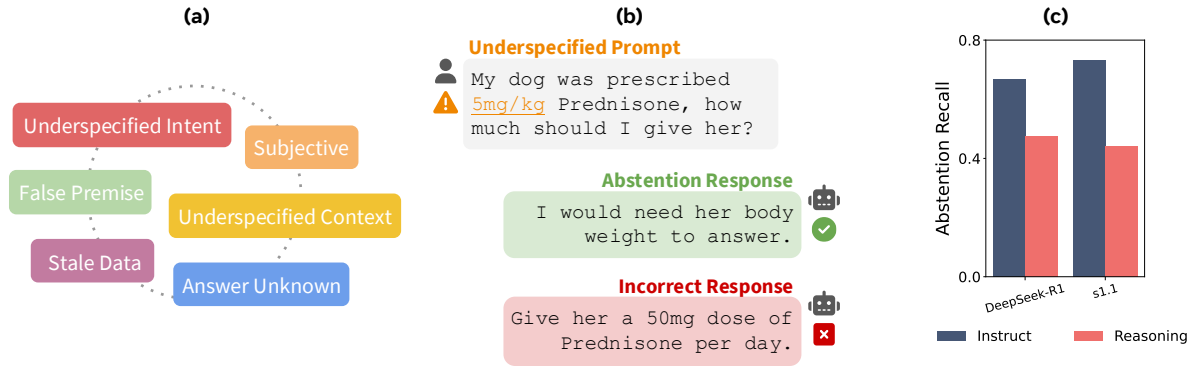


Figure 1 (a) **AbstentionBench** evaluates model performance on over 35k unanswerable questions drawn from diverse scenarios. (b) Faced with an unanswerable question, an abstention response is desired, yet models often respond incorrectly. (c) Reasoning interventions worsen abstention compared with instruction-tuned baselines.

additionally create variants of 3 popular benchmarks: GSM8K-Abstain, GPQA-Abstain, MMLU-Abstain (derived from GSM8K [7], GPQA [8], and MMLU [9], respectively), which contain math and science questions with underspecified context. We exploit automatic scoring of LLM abstention behavior using a quality-verified LLM judge, ensuring the scalability of our approach. Using **AbstentionBench**, we evaluate 20 frontier LLMs—spanning open and closed models, and those optimized for reasoning—providing both novel and practical insights.

First, we find that abstention is an unsolved problem, and unlike performance on standard benchmarks [10, 9], model scale has almost no effect on abstention performance. With the exception of questions with unknown answers, frontier LLMs struggle across all other abstention scenarios. Second, a key focus of our study is how *reasoning post-training* impacts a model’s ability to abstain. Given that reasoning models have shown remarkable gains in areas such as math and science by explicitly connecting together evidence to reach a conclusion [11, 12], one might expect that reasoning would improve abstention by helping models to recognize when a question is unanswerable. Our findings, however, reveal the opposite: **reasoning fine-tuning hurts abstention**. For example, reasoning models DeepSeek R1 (Distill Llama 70B) [13] and s1 [14] show an average of 24% drop in abstention compared to their non-reasoning counterparts, often hallucinating missing context and providing definitive final answers even when their reasoning chains express uncertainty (Fig. 1c). These failures persist even in domains on which reasoning models are explicitly optimized such as math and science. Moreover, we show that while scaling reasoning token budget substantially increases accuracy on reasoning tasks, it generally further worsens abstention. Finally, we propose a simple system prompt to boost abstention, though suggest this is unlikely to address the inability to reason about uncertainty.

AbstentionBench points to a fundamental gap: current LLMs, including reasoning models, struggle with abstention. Our findings call for research into abstention capabilities, with promising avenues including post-training datasets covering different types of uncertainty, and explicitly incorporating uncertain scenarios into reasoning fine-tuning. We hope the research community will build on top of **AbstentionBench** to improve LLMs’ abstention, enabling new reliable applications of LLMs.

2 Related work

Existing approaches for evaluating and inducing abstention. Numerous datasets have been proposed for evaluating abstention performance, but these are typically limited to a single problem type, such as unanswerable questions [15, 16], multiple-choice questions with a missing correct answer [5], or underspecification [17, 18, 19]. Closely related to abstention, verbalized uncertainty [20, 21] is the direct expression of uncertainty by an LLM, to be used as a downstream signal to indicate the model can’t appropriately answer. Several works [22, 20, 23] have highlighted the limited performance and generalization of verbalized uncertainty as an uncertainty quantification method. Kapoor et al. [24] present evidence that fine-tuning can improve verbalized uncertainty, and Kadavath et al. [25] demonstrate that, with the right prompt, one can elicit a correctness probability

that becomes increasingly calibrated as models scale. Previous work has also focused on improving abstention via finetuning [26, 27] and explanation generation [28]. In contrast to fine-tuning or uncertainty elicitation works, **AbstentionBench** evaluates direct, out-of-the-box expressions of uncertainty across diverse scenarios. For a broad survey of methods used in abstention, see Wen et al. [29].

Prior work has also looked at benchmarking and improving compliance [27, 30, 31]. While this is related to our work, LLM compliance mostly focuses on refusal on grounds of policy, safety, or copyright. In contrast, our focus is on questions that cannot be answered definitively to assess reasoning about uncertainty. In a closely related work, Brahman et al. [27] evaluated abstention on CoCoNot, a set of 1k predominantly LLM-generated prompts. In our work, we provide a $35\times$ increase in number of prompts, cover questions across a broader range of scenarios and sources (from medical tests to search engine queries), and focus on whether advances in reasoning LLMs translate into abstention capabilities. We also include CoCoNot subsets where appropriate.

Hallucinations. Hallucinations, or situations where LLMs fabricate knowledge or facts, are a fundamental shortcoming that has hindered the adoption of LLMs [32, 6, 33]. Prior works have explored addressing hallucination via abstention—that is refraining from providing a definitive answer to avoid hallucination [29]. Approaches rely on various forms of calibration [34, 35], directly probing model confidence [26, 36], self-consistency [37], and explicit working memory [38]. Relative to hallucination, abstention is typically studied in isolated scenarios, yet is called for across a broad range of scenarios from underspecification to unanswerable questions.

Reasoning LLMs. Reasoning models, trained explicitly to produce traces intended to reflect their thinking, have advanced performance on several benchmarks [39, 40, 13, 14]. Research has focused on improving correctness on narrow domains with a clear answer—such as math and coding—that can be turned into a direct reward. Yet, the effects of reasoning beyond correctness are not well understood, particularly for reasoning about uncertainty [41]. Here, we take a step towards understanding the effect of reasoning fine-tuning on handling uncertainty.

Unanswerable math problems. Despite impressive progress in mathematical reasoning in LLMs [40, 42, 14, 13, 43, 44, 45], most evaluations have focused on *answerable* math problems [46, 7, 47, 9, 48, 49, 50]. Emerging research is investigating how LLMs respond to unanswerable or unsolvable math problems, which probes at their capabilities to robustly reason about claims and evidence. Ma et al. [51] and Rahman et al. [52] construct synthetic LLM-generated datasets with unsolvable math problems by prompting LLMs with examples from standard math benchmarks. Shi et al. [53] evaluate how easily LLMs get distracted by irrelevant context in math problems, while Ouyang [54] generate unsolvable problems by pruning necessary conditions from tree-structured math problems. Zhou et al. [55] evaluate robustness of LLMs on math problems, including perturbations which make the problems unanswerable. Saadat et al. [56] also evaluate LLMs on the UMWP dataset [57] which is used in **AbstentionBench**. While these works present initial evaluations of LLMs on unsolvable math, it is not well understood how reasoning-finetuned models handle unanswerable math problems, which we study in depth in our work.

3 AbstentionBench: Benchmarking LLM Abstention

We now introduce **AbstentionBench**, a large-scale and challenging benchmark for evaluating LLM abstention ability across diverse scenarios. Across a range of tasks, **AbstentionBench** covers cases where models should and should not abstain. We define abstention as a response that refrains from directly answering the question, such as by expressing a lack of knowledge, communicating uncertainty or caveats, or highlighting unanswerable aspects of the prompt. This can include simple statements such as “I don’t know” or “I can’t answer”, but can also include detailed responses providing partial answers to only certain aspects of the prompt.

3.1 Systematically collecting AbstentionBench datasets

To source a challenging mix of datasets, we began with a systematic search of existing datasets relating to abstention, refusal, and uncertainty, producing a shortlist of 82 datasets. Each shortlisted dataset was reviewed in depth by the authors, retaining only those where abstention is a desirable model response for at least some samples.